AZAAN KHAN

Milton, Ontario

Computer Engineering | Embedded Systems, Hardware & Software

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Education

University of Waterloo — B.A.Sc. Computer Engineering (Co-op) Sep. 2026 – Apr. 2031 (Expected)

Waterloo, Ontario | Coursework: Digital Systems, Programming, Calculus, Linear Algebra, Engineering Physics, ECE 198 Design Studio

Technical Skills

- **Languages / Web:** C, C++, Python, JavaScript, TypeScript, React, Node.js, Express.js, Vite
- **Embedded / Electrical:** ESP32, STM32, Arduino IDE, STM32CubeIDE, Altium Designer, circuit/PCB design, soldering, wiring, sensor integration
- **Tools / Design:** GitHub, VS Code, command line, SolidWorks, CAD/CAM, 3D printing, debugging, technical documentation

Projects

ECE 198 Design Studio — Interactive Rehabilitation Device Sep. 2026 – Present

University of Waterloo | Team engineering design project | In progress

- Collaborating on a compact, motion-based activity device intended to encourage participation in exercise among people experiencing muscle loss, including cancer-related sarcopenia.
- Developing Arduino/ESP32-based interaction concepts, exploring sensor data and device communication for movement-driven mini-games.
- Tested and incorporated many troubleshooting phases for the pule oximeter and accelerometer.
- Planning a web-connected leaderboard to present activity data; researching physical and motivational barriers to exercise to inform design decisions.

ClinTempo — AI-Assisted EMR Chrome Extension Sep. 2026

React, TypeScript, Vite, Node.js, Express.js, Corti AI API

- Built a clinical documentation extension integrating an AI medical-scribe API with structured, customizable consultation-note templates.
- Developed frontend and backend workflows to produce EMR-ready notes, focusing on usability, data flow, and debugging.
- Incorporated feedback from licensed Canadian family physicians to refine the interface and documentation workflow.
- Reduced clinical documentation typing time for family physicians by 25 percent.

ESP32 Washing Machine Notifier (Aug. 2026)

bullets

Designed an accelerometer-based monitoring circuit; wrote C++ firmware and integrated ntfy mobile alerts.

Smartphone Repair & Restoration (Aug. 2026)

Replaced a damaged display and used command-line diagnostics to restore device software.

Vernier Calipers Holder (Dec. 2025)

Converted CAM/DXF geometry into a SolidWorks model and prepared STL/G-code for PLA 3D printing.

Professional Experience

BioMechatronics (Biotron) Design Team — Software & Electrical Member Sep. 2026 – Present

University of Waterloo | Robotic exoskeleton project

- Contributing to an assistive exoskeleton design focused on mobility support for people experiencing muscle loss.
- Designed battery-pack and motor-control circuit concepts in Altium Designer; supported electrical wiring and PCB soldering.
- Developed and debugged STM32 control logic, testing hardware/software interactions and reviewing design changes with teammates.

EMR Support Analyst (Part-Time) — Intrepid Health Group Jun. 2026 – Aug. 2026

- Maintained confidential patient information, drafted notes under physician supervision, and coordinated appointment and clinical communications.

Pharmacy Assistant — Whole Health Pharmacy Group Jul. 2025 – Aug. 2025

- Supported prescription packaging, inventory tracking using pharmacy software, and customer service with attention to accuracy.

IT Technician (Apprentice) — AESSEAL Jun. 2024

- Assisted with hardware/software troubleshooting, computer setup, network checks, and exposure to SAP ERP workflows.

Leadership & Recognition

Waterloo EV Club: Social Media / Manufacturing | **Footy4Hope:** Helped raise CAD $1,100+ | **Relay For Life:** Committee

Awards: Junior British Physics Olympiad Silver (2024); Wickersley Scholar Award (2025); Platinum Academic Effort Award, 1st of 345 (2025)